

State-of-Polarisation (SoP) Monitoring of Wind-Induced Resonance Oscillations on Power Lines

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Abstract *Aeolian vibrations cause fatigue damage to power-line infrastructure. We monitor SoP on a live OPGW link over seven months, correlating with wind-speed data. Discrete resonance tones below 25 Hz occur at wind speeds up to 7 m/s, demonstrating SoP sensing for proactive structural condition monitoring. ©2026 The Authors*

Introduction

Power-line infrastructure requires continuous structural health monitoring to prevent fatigue-related failures. Fibre sensing using optical fibre embedded in optical ground wire (OPGW) offers a monitoring approach along the entire infrastructure. SoP sensing is shown to be sensitive to mechanical perturbations on aerial links while requiring low-cost monitoring hardware [1].

SoP variations in aerial fibres caused by wind and power-line current components around 50 Hz have been previously reported [2]. Sensing over a 524 km live aerial OPGW fibre route showed that the SoP response is dominated by strong $n \times 50$ Hz components, while lower-frequency variations relate to environmental effects [3]. Measurements on a 6 km fibre spun on one of the power wires on an aerial high-voltage link demonstrated that low-frequency SoP spectral properties correlate with wind conditions, also on shorter links [4].

Wind-induced motion in power lines causes a variety of vibration effects, including aeolian vibration, galloping, and, for bundled phase conductors, subspan oscillations [5–7]. Subspan oscillation is a wake-induced flutter-type instability caused by the aerodynamic shielding of leeward subconductors by windward ones, and it typically occurs in the 0.5–2 Hz range [6]. For a single OPGW span, however, aeolian vibration is generally the dominant mechanism. This type of vibration arises from vortex shedding and produces narrowband standing-wave behaviour at frequencies that depend on wind speed, cable diameter, span length, and tension, with a frequency range of 3–100 Hz for wind speeds of 0.5–7 m/s [5,7]. Above these wind speeds, the vortices become more turbulent, reducing the wind energy input to the standing wave in the line [8]. When the vortex-shedding frequency approaches the span's mechanical natural frequency, lock-in occurs and the shedding and oscillation frequencies converge to a single value near the span mode [5]. In this lock-in behaviour, [7] models steady-state aeolian vibration as a standing-wave response at the lock-in frequency. To minimise vibrations,

dampers are widely studied and used to dissipate oscillating frequencies that could otherwise cause damage to the infrastructure [7].

Mechanical cable motions have been reported to produce SoP fluctuations at the same frequencies as the mechanical excitations themselves [8], however, the relationship between SoP spectral content and wind-induced vibration effects over long field-deployed live OPGW links has not been previously explored in detail.

In this paper, we perform a long-duration field trial using SoP sensing on a live 86 km OPGW link, identifying different vibration effects and potentially harmful vibrations. We identify discrete frequency tones and spectral energy bands that correlate with concurrent wind-speed measurements. The dataset spans seven months, including a 24-day de-energised period (i.e. no current through the line), enabling observations of wind-induced SoP variations also without the impact from the 50 Hz power-frequency and its harmonics.

Field Test Setup

The field trial was conducted in cooperation with Svensk Kraft on a live 400 kV power line in Sweden, as illustrated in Fig. 1. The sensing path comprised an 86 km OPGW span looped at the remote end, yielding a 172 km round-trip. A continuous-wave optical signal, amplified by an Erbium-Doped Fibre Amplifier (EDFA) booster, was launched into the fibre at +17 dBm, looped at the far end and received back at the originating end by the SoP sensing instrument at -24 dBm after filtering and attenuation, ensuring a power level below the -20 dBm maximum allowable power to the instrument. Hourly wind-speed data over the seven-month trial were taken from a weather station close to the loop-back site [9].

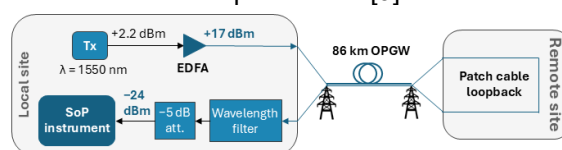


Fig. 1: Experimental field-trial setup on an 86 km OPGW line.

The sensing instrument is Polarisation Beam Splitter (PBS)-based, and measures the S_1 Stokes parameter, previously shown to be sufficient for detecting mechanical impacts on fibre cables [10]. The receiver is AC-coupled, suppressing DC components and low-frequency drifts, improving sensitivity to wind-induced frequencies.

On an OPGW, the 50 Hz power frequency and its harmonics appear as strong SoP components [3] and carry intermodulation products around these tones caused from mixing with direct wind-induced frequencies [2]. The dataset includes 24 days during which the line was de-energised for maintenance, providing a reference condition without power-frequency contributions. This enables evaluation of spectral overlap between the electrical-current-related components and wind-induced SoP variations [2]. Together with the overall seven-month duration SoP measurement period, the dataset enabled a comprehensive investigation of the SoP spectral response and the identification of potentially harmful wind-induced oscillations.

Results

First, a comparison is performed for both energised and de-energised periods, exploring how the 50 Hz tone, its overtones and their sidebands interfere with the direct wind-induced SoP changes. Fig. 2 shows two heatmaps of the de-energised (a) and energised (b) periods. The plots show SoP Power Spectral Density (PSD) up to 50 Hz, using 1 h averaging windows, as a function of wind speed, with resolutions of 0.2 Hz and 0.1 m/s respectively.

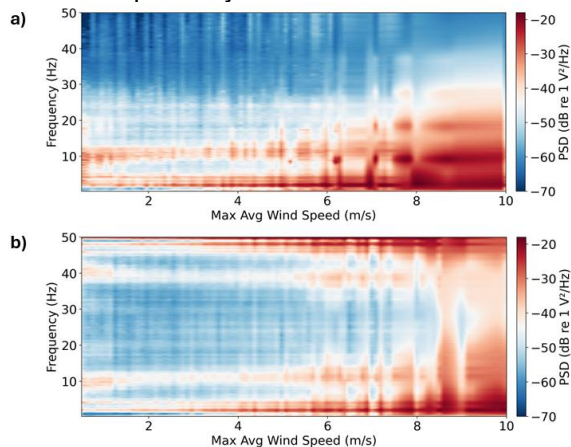


Fig. 2: Heatmaps of the 24-day de-energised period (a) and a 30-day energised period (b) showing PSD for frequency against wind speeds.

The SoP response during the de-energised period of Fig. 2a shows increased energy and higher frequency components for increasing wind speeds. Frequency components above 25 Hz, where aeolian vibrations could be expected, are

of low energy, even for strong wind speeds. This indicates that the installed dampers on the infrastructure are preventing high-frequency oscillations from happening and from causing fatigue of the infrastructure. Furthermore, the figure shows that, because of the low intensity in frequency bands above 25 Hz, the analysis of the energised OPGW can be performed without significant interference from the 50 Hz. Fig. 2b illustrates how the 50 Hz tone mixes with the directly wind-induced low frequency tones, creating a mirroring effect around 25 Hz potentially complicating the identification of any possible direct wind-induced frequency tone above this limit, if present.

From both heatmaps showing multi-day averaging, two distinct frequency bands are visible at 1–3 Hz and 9–12 Hz. This shows that, in average, these two bands present the highest power spectral density. However, wind-induced vibrations are more complex than these two bands suggest. The spans of the measured high-voltage power lines are in the range of 300 m in length, and the 86 km link comprises roughly 286 spans, each with slightly different length, tension, and wind exposure.

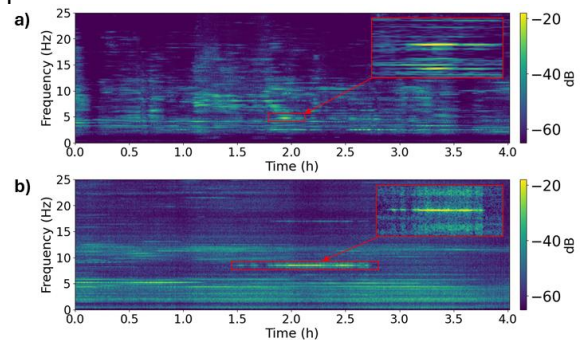


Fig. 3: Two four-hour-period spectrograms. Wind speeds of (a) 1 m/s and (b) 4 m/s illustrating examples of periods where resonance oscillations can be observed.

Fig. 3 shows two four-hour spectrograms exemplifying the frequency fingerprint of SoP variations recorded on the line at two different wind speeds. At wind speeds of 1 m/s, Fig. 3a exhibits several short-term distinct tones of varying intensity and duration, at frequencies ranging from 2 to 20 Hz. Fig. 3b, for wind speeds of 4 m/s, presents a more diffuse spectrum with wider frequency bands, and a highlighted 8.5 Hz tone with a harmonic at 17 Hz, with a duration of close to an hour. From the frequencies of these observations, we classify these tones as span-specific resonance phenomena caused by vortex-shedding winds [5,7].

To further analyse the occurrence of these narrowband short-term tones over longer periods, Fig. 4 shows PSDs of short-term windows across a full month, grouped into three wind-speed bins spanning from 0.5 to 7.5 m/s.

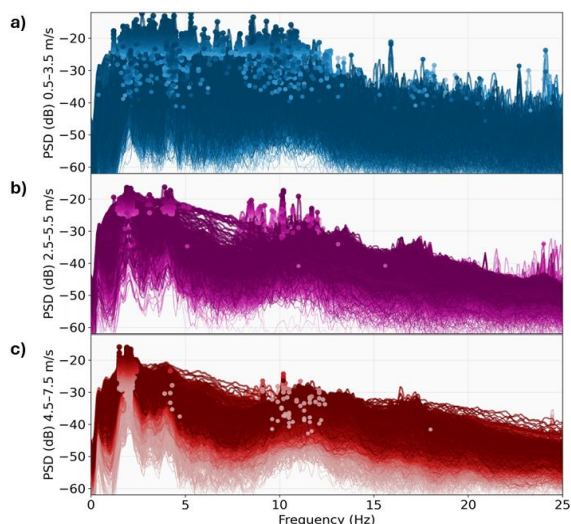


Fig. 4: PSD contours of individual 10-minute windows at 0–25 Hz for the energised line. Wind speeds in the ranges of (a) 0.5–3.5 m/s, (b) 2.5–5.5 m/s, and (c) 4.5–7.5 m/s

For 30 days of data on the line energised, the PSDs of 10-minute time windows are shown up to 25 Hz, calculated with 0.1 Hz frequency resolution. Three wind speed bins of 0.5–3.5 m/s (low winds), (a), 2.5–5.5 m/s (b) (middle winds), and (c) 4.5–7.5 m/s (strong winds) were chosen to represent the different vibrational states of the line. Winds stronger than 7.5 m/s are not shown, as the presence of resonant oscillations above these speeds decrease substantially. The PSDs for each of the 10-minute time windows are drawn in the figures as single lines. For highlighting the occurrence of frequency peaks with substantially higher values (100 times) than the average of the corresponding frequency window, dots are placed to indicate values with a higher peak-to-mean level than 20 dB over the mean PSD of each window.

At the lowest wind speeds, Fig. 4a shows that the occurrence of peaks is most prominent, indicating that the highest prevalence of resonant oscillations in the infrastructure is for the lowest wind-speeds. These peaks occur mainly at frequencies between 2 and 12 Hz. For middle wind speeds in Fig. 4b, the abundance of peaks decreases, and for the stronger wind speeds plotted in Fig. 4c, almost all distinct single tones disappear. This is indicative of more turbulent cable motion outside the vortex-induced lock-in regime. At these strongest wind speeds, what remains is a prominent band around 2 Hz, which may correspond to wake-induced vibrations in bundled phase conductors transferred through the towers.

Lastly, Fig. 5 extends the analysis of shorter duration discrete tones to six months of data on the energised line. Building on the 10-minute windows of Fig. 4, we select the 1000 windows with the highest PSD peaks and average the standard

deviations of their PSDs. Since resonant oscillations occur intermittently rather than continuously, restricting the average to the windows with the most prominent tones prevents quiet periods from washing out the signature in the overall average. This provides an overview of the full dataset, confirming that the most prominent SoP variations, and therefore the largest resonant motions in the line, occurs for the lowest wind speeds.

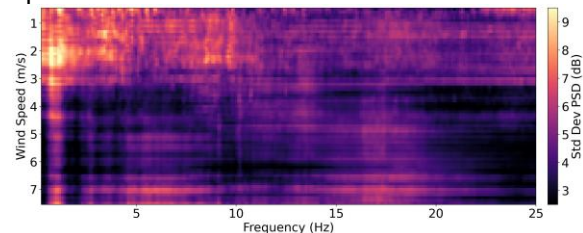


Fig. 5: Heatmap of the average standard deviation of the top 1000 10-minute windows' PSDs with highest peaks. Frequency spanning 0–25 Hz and wind speeds 0.5–7.5 m/s

Conclusions

We demonstrated how SoP monitoring can be applied for identifying potentially harmful cable oscillations in aerial power-line infrastructures. On a live 86 km OPGW link, wind-induced vibrations are analysed in a seven-month field trial. Discrete frequency tones below 25 Hz, predominantly between 2–12 Hz, were identified at wind speeds below 7 m/s, consistent with aeolian vibration driven by vortex-shedding lock-in. The 50 Hz power-frequency and its overtones give dominant signals when monitoring SoP on power-lines. A 24-day de-energised window allowed observations avoiding power-related interference, confirming the wind-induced origin of observed frequency signatures during the energised period, and validating the applicability of the approach for energised lines. Strong vibrations above 25 Hz were not observed, indicating the correct functioning of dampers. At stronger wind speeds, the spectral response broadened, making individual tones less distinct, indicating turbulent motion outside the lock-in regime. The strongest tonal activity occurred, however, at the lowest wind speeds, where more laminar flow promotes stronger vortex shedding.

These results show that SoP sensing on live OPGW can reveal more than just the presence of wind loading, as previously reported. Specific cable-vibration regimes are found, supporting proactive identification of potential fatigue-inducing aeolian vibrations. Future work may relate observed tones to individual span parameters and extend the analysis to sub-hertz frequencies for observing potential low-frequency high-amplitude motions.

Acknowledgements

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References

- [1] S. Bjørnstad, J. Ali, T. Dreibholz, and E. Sæthre, "A Scalable Data Collection System for Continuous State of Polarisation Monitoring," in 2023 23rd International Conference on Transparent Optical Networks (ICTON), Bucharest, Romania, 2023, pp. 1-4.
- [2] J. Wuttke, P. M. Krummrich, and J. Rosch, "Polarization oscillations in aerial fiber caused by wind and power-line current," *IEEE Photonics Technology Letters*, vol. 15, no. 6, pp. 882-884, 2003. DOI: <https://doi.org/10.1109/LPT.2003.811143>
- [3] M. Mazur, D. Wallberg, L. Dallachiesa, E. Börjeson, R. Ryf, M. Bergroth, B. Josefsson, N. K. Fontaine, H. Chen, D. T. Neilson, J. Schröder, P. Larsson-Edefors, and M. Karlsson, "Field Trial of FPGA-Based Real-Time Sensing Transceiver over 524 km of Live Aerial Fiber," in Optical Fiber Communication Conference (OFC), San Diego, CA, USA, 2023, paper Tu3G.4. DOI: <https://doi.org/10.1364/OFC.2023.Tu3G.4>
- [4] K. S. Y. Skarvang, S. Bjørnstad, E. Sæthre, and D. R. Hjelme, "Local Wind Impact Sensing using State of Polarization Measurement on a Live Short-Haul Aerial Fibre Cable," in Optical Fiber Communication Conference (OFC), San Diego, CA, USA, 2024, paper Tu2J.5. DOI: <https://doi.org/10.1364/OFC.2024.Tu2J.5>
- [5] A. R. E. Oliveira and D. G. Freire, "Dynamical modelling and analysis of aeolian vibrations of single conductors," *IEEE Transactions on Power Delivery*, vol. 9, no. 3, pp. 1685-1693, 1994. DOI: <https://doi.org/10.1109/61.311193>
- [6] J. Wang, "Overhead Transmission Line Vibration and Galloping," in 2008 International Conference on High Voltage Engineering and Application (ICHVE), Chongqing, China, 2008, pp. 120-123. DOI: <https://doi.org/10.1109/ICHVE.2008.4773888>
- [7] D. F. Campos, A. E. Ajras, L. G. Goytíño, and M. T. Piován, "Uncertainties Propagation and Global Sensitivity Analysis of the Aeolian Vibration of OPGW Cables," *Journal of Verification, Validation and Uncertainty Quantification*, vol. 8, no. 1, Art. no. 011004, 2023. DOI: <https://doi.org/10.1115/1.4056976>
- [8] R. Roberge, "Case Study: PMD Measurement on Aerial Fiber under Wind-Induced Oscillations and Vibrations," EXFO Technical Note 039, 2009.
- [9] SMHI, "Ladda ner väderobservationer," SMHI. [Online]. Available: <https://www.smhi.se/data/hitta-data-for-en-plats/ladda-ner-vaderobservationer>. [Accessed: 24-Feb-2026].
- [10] S. Bjørnstad, A. Doval, and A. Tysdal, "First observations of subsea cable physical contacts combining State of Polarization (SoP) and Distributed Acoustic Sensing (DAS)," in Optical Fiber Communication Conference (OFC), Los Angeles, CA, USA, 2026.